

CSA0708 COMPUTER NETWORKS

DONE BY
S. DHANUSH
192524317



SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

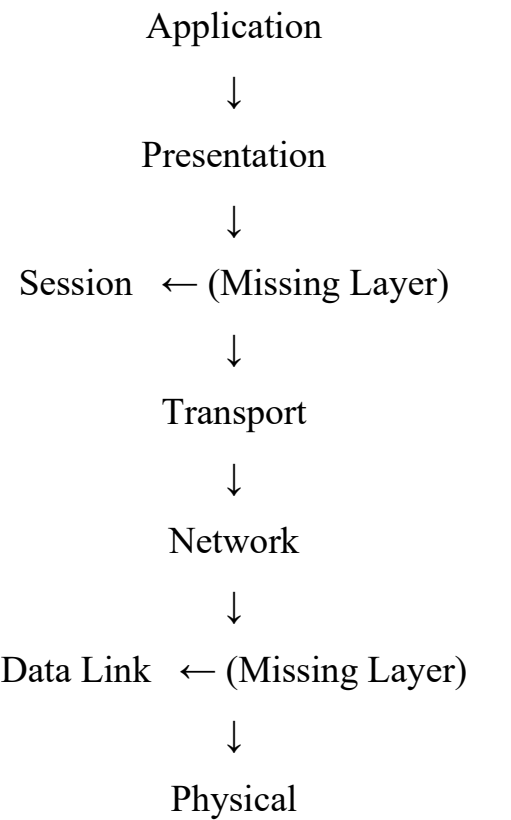
CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A
CONCEPT MAPPING

1. A partially completed concept map is given below:
Application → Presentation → ___ → Transport → Network → ___ →
Physical. Analyze the missing layers and explain how their functions are
essential for reliable communication.

Concept Map



Explanation

The two missing layers are:

1. Session Layer (Layer 5)
2. Data Link Layer (Layer 2)

Functions of Each Layer

Application Layer (Layer 7)

- Provides network services to users.
- Examples: Web browser, Email, FTP.

Presentation Layer (Layer 6)

- Translates data into a common format.
- Performs encryption, decryption, and compression.

Session Layer (Layer 5) (*Missing*)

- Establishes, manages, and terminates communication sessions.
- Maintains synchronization between devices.
- Allows interrupted communication to resume.

Transport Layer (Layer 4)

- Provides reliable end-to-end communication.
- Performs segmentation, error recovery, and flow control.
- Uses TCP or UDP.

Network Layer (Layer 3)

- Determines the best path for data.
- Uses IP addresses for routing packets.

Data Link Layer (Layer 2) (*Missing*)

- Transfers frames between directly connected devices.
- Detects errors using CRC.
- Uses MAC addresses.

Physical Layer (Layer 1)

- Transmits bits over cables or wireless media.
- Defines electrical and mechanical specifications.

Analysis: Why are the Missing Layers Essential?

- The Session Layer ensures communication sessions are created, maintained, and properly closed. Without it, long communications may fail or become unsynchronized.
- The Data Link Layer ensures error-free communication between neighboring devices by detecting transmission errors and using MAC addresses for local delivery.

- Together, these layers improve reliability, accuracy, and efficient communication across the network.

Final Answer (Exam Format)

Missing Layers:

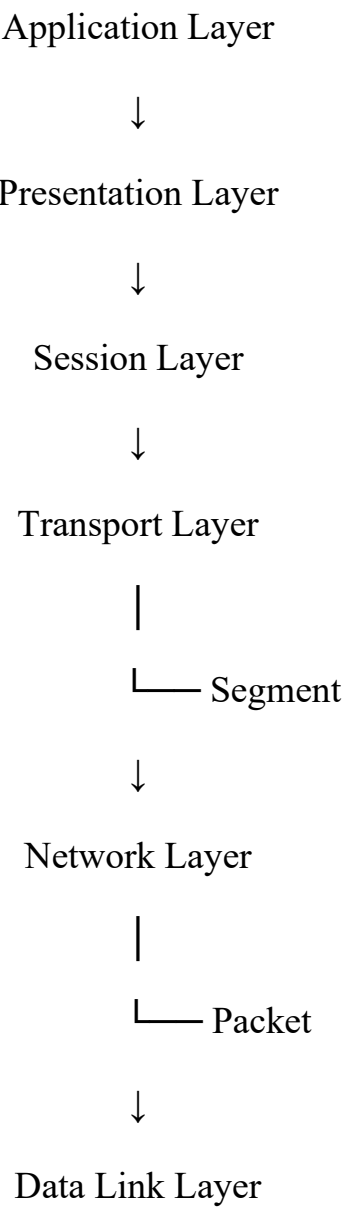
- Session Layer
- Data Link Layer

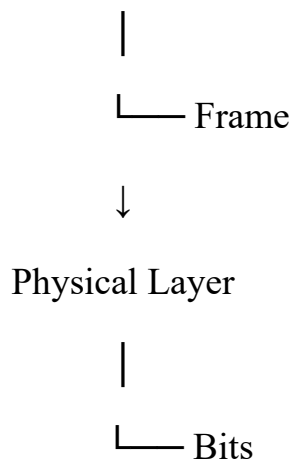
Conclusion:

The Session Layer manages communication sessions, while the Data Link Layer provides reliable local data transfer and error detection. These layers are essential because they maintain synchronization, detect transmission errors, and ensure reliable communication between devices.

2. **Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.**

Concept Map





Relationship Between OSI Layers and Data Units

OSI Layer	Data Unit	Function
Transport Layer	Segment	Divides data into smaller segments and ensures reliable delivery.
Network Layer	Packet	Adds IP addresses and determines the best route to the destination.
Data Link Layer	Frame	Adds MAC addresses and error detection information for local delivery.
Physical Layer	Bits	Converts data into binary bits (0s and 1s) and transmits them through the transmission medium.

Analysis: How Transformation of Data Units Supports Transmission

- The Transport Layer divides the original message into segments for efficient transmission.
- The Network Layer encapsulates each segment into a packet by adding IP addresses, allowing routing across networks.
- The Data Link Layer converts packets into frames by adding MAC addresses and error detection information, ensuring reliable communication between directly connected devices.

- The Physical Layer converts frames into bits (0s and 1s) and transmits them through cables, fiber optics, or wireless media.
- At the receiver's end, the process is reversed (decapsulation), where bits become frames, frames become packets, packets become segments, and finally the original data is reconstructed.

Conclusion

The transformation of data from Segment → Packet → Frame → Bits is called encapsulation. Each OSI layer adds its own control information, ensuring accurate addressing, routing, error detection, and reliable communication. At the receiving end, decapsulation restores the original data, enabling successful end-to-end communication.

3. Given the concept map:

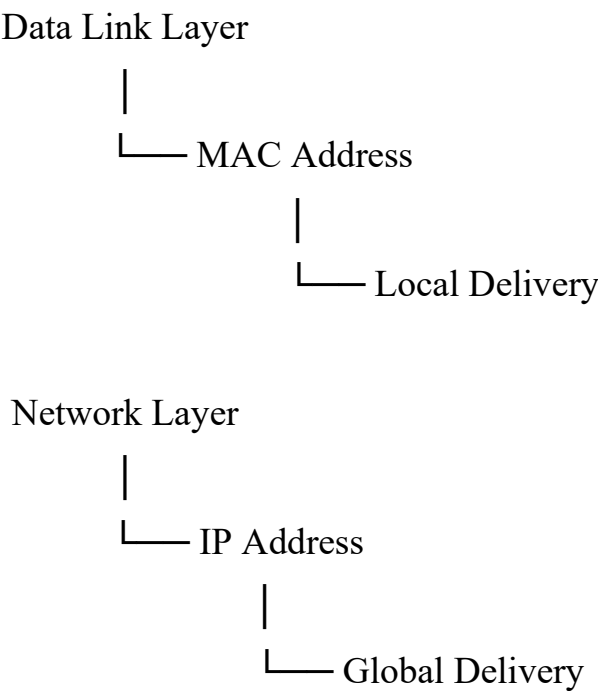
Data Link → MAC Address → Local

Delivery Network → IP Address

→ Global Delivery

Explain how these relationships ensure complete data delivery across networks.

Concept Map



Explanation

Data Link Layer → MAC Address → Local Delivery

- The Data Link Layer uses MAC (Media Access Control) addresses.
- A MAC address is a unique physical address assigned to a network interface card (NIC).
- It ensures data is delivered correctly within the same local network (LAN).
- Switches use MAC addresses to forward frames to the correct device.

Network Layer → IP Address → Global Delivery

- The Network Layer uses IP (Internet Protocol) addresses.
- IP addresses uniquely identify devices across different networks.
- Routers use IP addresses to determine the best path and forward packets to the destination.
- This enables communication over the Internet or between different networks.

Analysis: How These Relationships Ensure Complete Data Delivery

- The Network Layer first uses the IP address to route packets from the source network to the destination network (global delivery).
- Once the packet reaches the destination network, the Data Link Layer uses the MAC address to deliver the frame to the correct device (local delivery).
- Thus, IP addresses provide end-to-end routing between networks, while MAC addresses ensure accurate delivery within the local network.
- Together, these layers guarantee that data reaches the correct destination efficiently and reliably.

Conclusion

The Network Layer ensures global delivery using IP addresses, while the Data Link Layer ensures local delivery using MAC addresses. The combination of these two addressing methods enables complete and reliable communication across both local and worldwide networks.

4. A concept map shows:

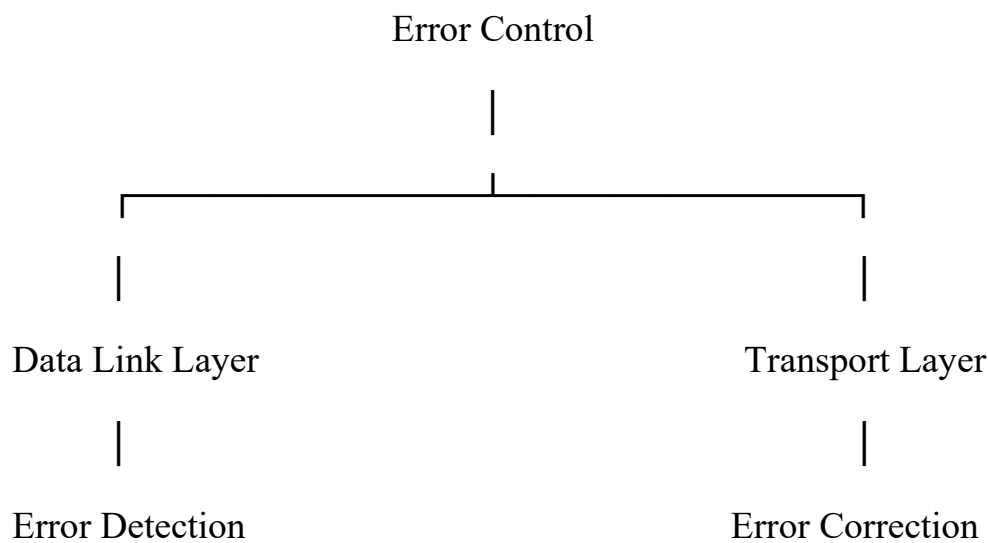
ErrorControl

→**DataLinkLayer**→**ErrorDetection**

→ **Transport Layer** → **Error Correction**

Analyze how the interactionbetween these layers improves reliability in data transmission.

Concept Map



Explanation

Data Link Layer → Error Detection

- Detects errors that occur during data transmission between two directly connected devices.
- Uses techniques such as CRC (Cyclic Redundancy Check) and checksums.
- If an error is detected, the corrupted frame is discarded or a retransmission is requested.

Transport Layer → Error Correction

- Provides reliable end-to-end communication.
- Uses TCP to detect lost or corrupted segments.
- Performs retransmission, sequencing, acknowledgments (ACKs), and flow control to ensure all data reaches the destination correctly.

Analysis: How These Layers Improve Reliability

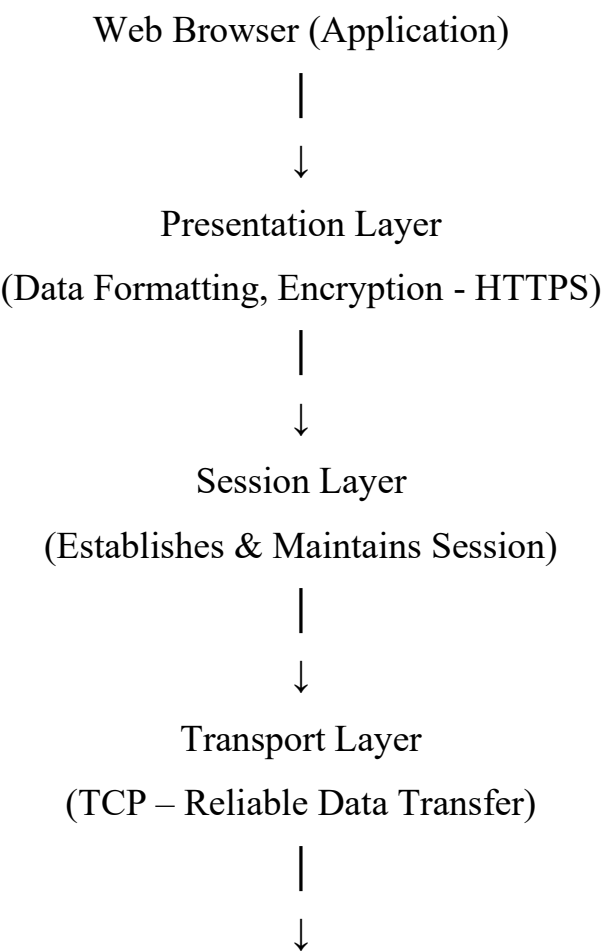
- The Data Link Layer detects transmission errors on a single network link before forwarding data.
- The Transport Layer ensures that the complete message is delivered correctly from the source to the destination, even if packets are lost or corrupted during transmission.
- If the Data Link Layer cannot recover from an error, the Transport Layer requests retransmission of the missing or damaged data.
- The interaction of these two layers minimizes data loss, improves accuracy, and provides reliable communication across the network.

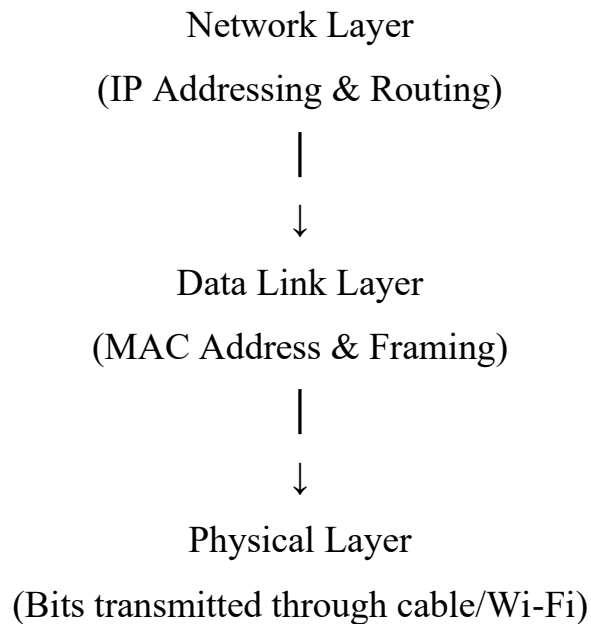
Conclusion

The Data Link Layer improves reliability by detecting errors during local transmission, while the Transport Layer provides end-to-end error correction and retransmission. Together, they ensure accurate, reliable, and efficient data communication.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

Concept Map





Explanation

Application Layer

- The user enters a website URL (e.g., www.google.com) in the web browser.
- HTTP/HTTPS is used to request the webpage.

Presentation Layer

- Formats the data and encrypts/decrypts it using HTTPS (SSL/TLS).

Session Layer

- Establishes, manages, and terminates the communication session between the browser and the web server.

Transport Layer

- Uses TCP to divide data into segments and ensures reliable delivery using acknowledgments and retransmissions.

Network Layer

- Uses IP addresses to route packets from the user's device to the web server.

Data Link Layer

- Uses MAC addresses to deliver frames within the local network and performs error detection.

Physical Layer

- Converts frames into bits and transmits them through cables, fiber optics, or Wi-Fi.

Analysis: Effect of Failure in One Layer

- If the Physical Layer fails, no data can be transmitted.
- If the Data Link Layer fails, frames cannot be delivered correctly within the local network.
- If the Network Layer fails, packets cannot reach the destination because

routing fails.

- If the Transport Layer fails, reliable communication is lost, causing incomplete or corrupted webpage loading.
- If the Session Layer fails, the connection between the browser and the server is interrupted.
- If the Presentation Layer fails, encrypted data cannot be decoded correctly.
- If the Application Layer fails, the browser cannot display the requested webpage.

Since every OSI layer depends on the services of the layer below it, a failure in any one layer can disrupt the entire web browsing process.

Conclusion

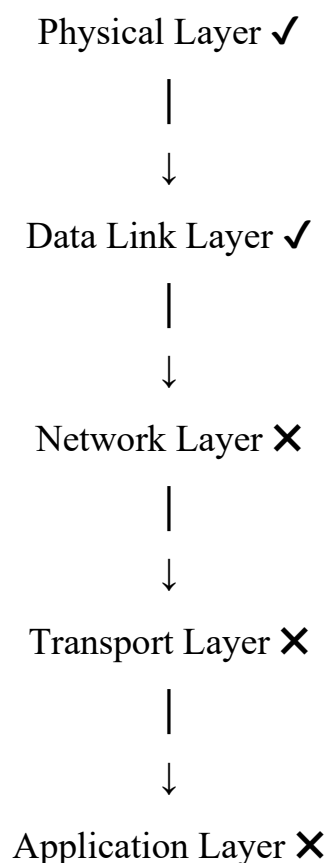
Web browsing requires all seven OSI layers to work together. Each layer performs a specific function, from creating the web request to transmitting data as bits.

Failure in any layer interrupts communication, making it impossible to access the requested webpage.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Concept Map



Explanation

Physical Layer ✓

- The physical connection (cables, Wi-Fi, or signals) is working correctly.
- Data can be transmitted between devices.

Data Link Layer ✓

- Frames are successfully transferred within the local network.
- MAC addressing and error detection are functioning properly.

Network Layer ✗

- The Network Layer has failed.
- IP addressing or routing is not working, so packets cannot reach the destination network.

Transport Layer ✗

- Since the Network Layer cannot deliver packets, the Transport Layer cannot establish reliable end-to-end communication.

Application Layer ✗

- Applications such as web browsers, email, or file transfer cannot communicate because the lower layers have failed.

Analysis: How the Concept Map Helps in Troubleshooting

- The concept map shows that the Physical and Data Link layers are functioning correctly, so the problem is not with the cable, Wi-Fi, or local network.
- The first failed layer is the Network Layer, indicating that the issue is likely related to IP addressing, routing, or router configuration.
- Because higher layers depend on the Network Layer, the Transport and Application layers also fail.
- By identifying the first failed layer, network administrators can focus on troubleshooting the Network Layer instead of checking every layer, saving time and improving fault diagnosis.

Conclusion

The concept map indicates that the Network Layer is the source of the problem. Since the OSI layers depend on one another, failure at the Network Layer prevents the Transport and Application layers from functioning. Concept maps help identify the faulty layer quickly, making network troubleshooting faster and more efficient.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation